

**LAMBA**

# A digital twin for a car

Android app, FastAPI backend, PostgreSQL timeline, and AI-assisted chat for structured vehicle records.

## PROBLEM

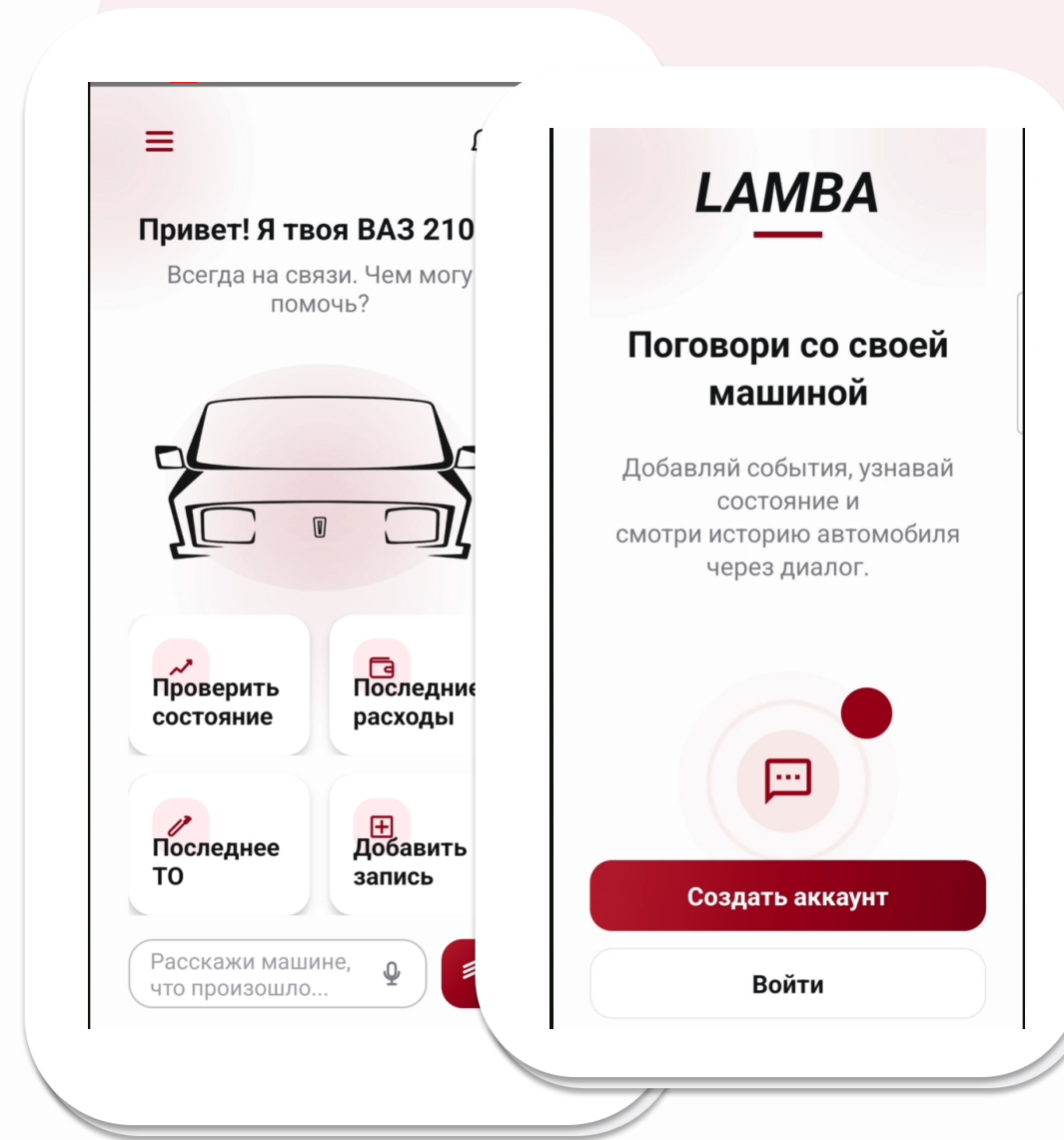
vehicle data loss

## SOLUTION

AI-twin for the car

## END-USERS

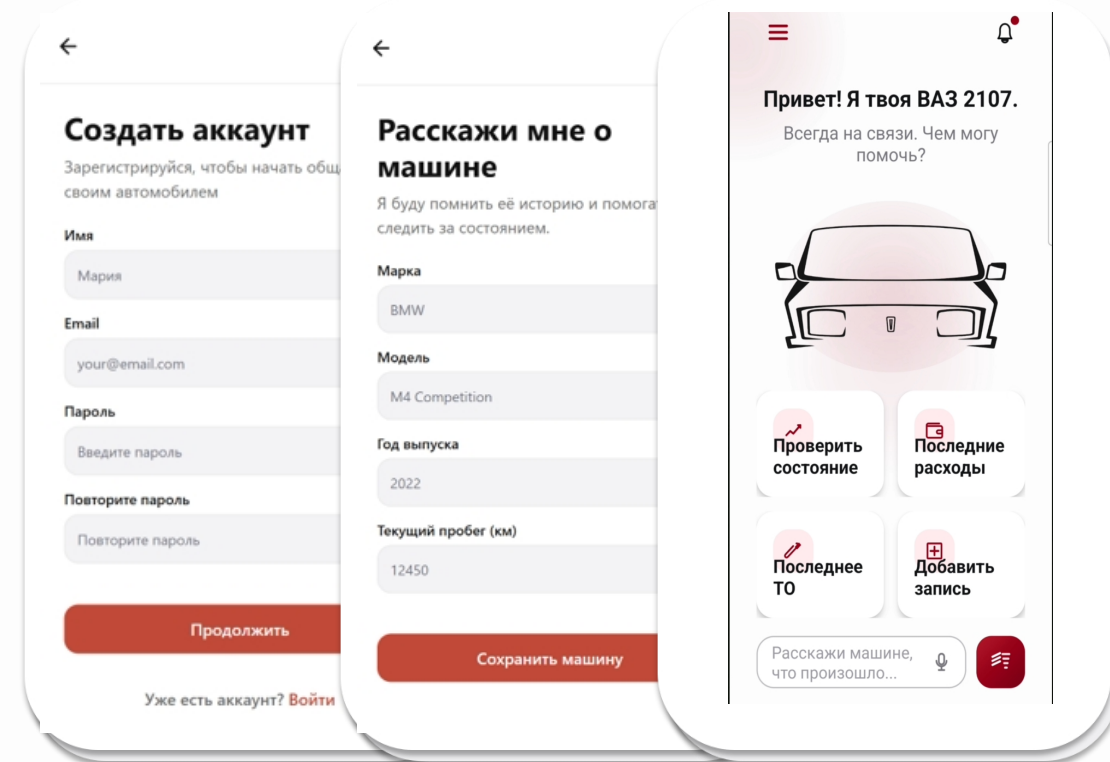
car owners



## Show the product flow first.

Video demo

<https://drive.google.com/drive/folders/1r27kEbYegLTjaM85RWaCVcfKkgIq-NO9>



register

vehicle

chat

1

### Register

Create account or log in.

2

### Vehicle

Add your car profile.

3

### Chat

Send natural message.

4

### Timeline

Review structured record.

## TEAM

# Responsibilities map directly to the increment.

Each role owns a visible part of the MVP evidence.

VT

**Vasilisa Tumakina**

Team lead / backend

Planning, backend coordination, release evidence, AI integration, chat parser

EI

**Elizaveta Ionina**

Frontend developer

Onboarding, login/register integration

MC

**Maria Chizhikova**

Frontend developer, UI/UX

Vehicle/profile screens and UI flow support

AK

**Amira Khaliullova**

Backend developer

Auth, vehicle endpoints, validation/tests

IG

**Ivan Grishaev**

AI engineer

Chat parser, AI integration, event extraction

## REQUIREMENTS

# MVP scope is clear, quality gates stay visible.

The deck separates must-have delivery from later expansion.

### Must have

- Registration
- Add vehicle
- Chat messages
- Records from chat
- Timeline
- Assistant
- Basic statistics

### Should / could

- Maintenance recommendations
- Notifications
- Voice messages
- Repair receipts

### Out of MVP

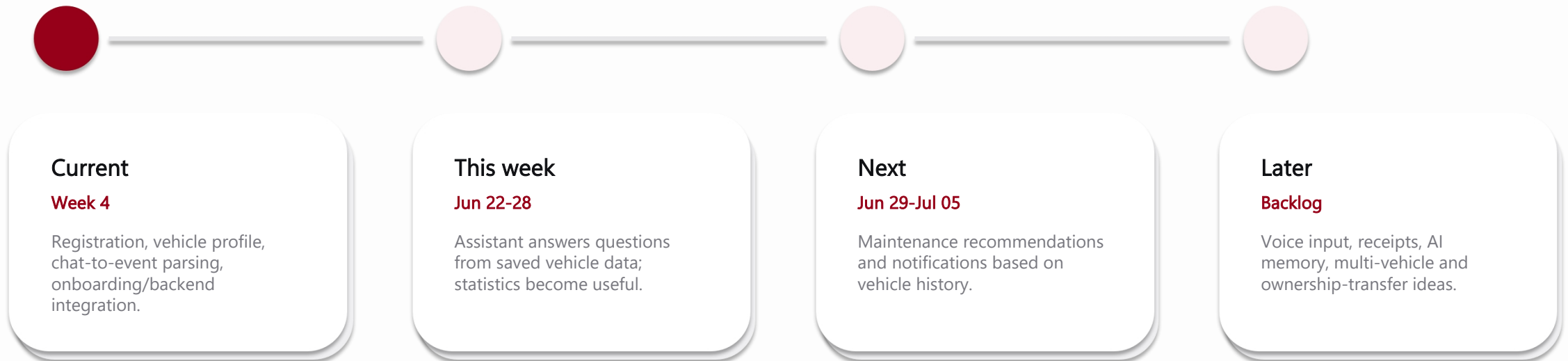
- Multi-vehicle management
- Ownership transfer
- OBD-II integration

### Quality

- No plain-text passwords
- User-specific backend data
- Automated tests
- Review, changelog, docs

## Stabilize the core flow before expanding AI behavior.

The next work is sequenced around user value and delivery risk.



## Customer feedback is already shaping scope.

The team is keeping MVP decisions small, usable, and demoable.

### Already usable

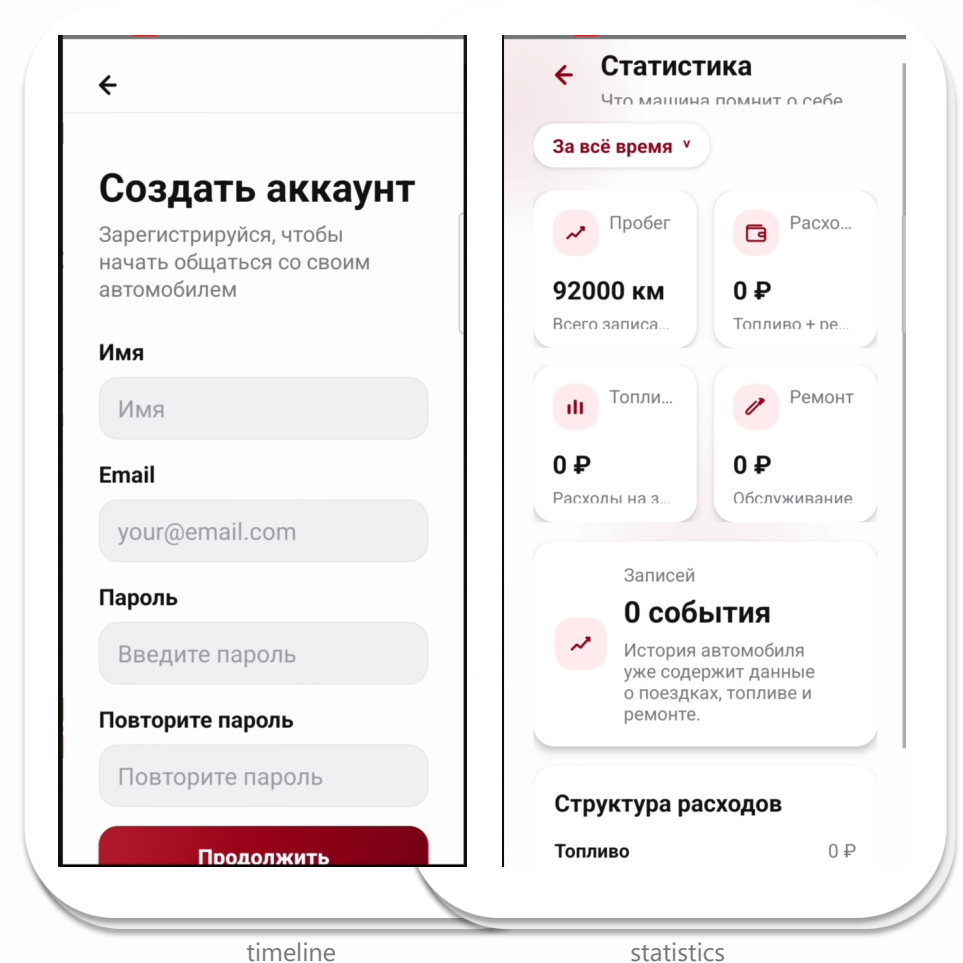
Deployed backend and Swagger · Repeatable smoke-check flow · Android screens for core flow

### Useful feedback

Login + password is enough · Manual brand/model input is acceptable · One vehicle per user is acceptable

### Current challenges

Password security hardening · User-specific data verification · Keep advanced AI outside MVP



## Repository and deployed API

GitHub repository

<https://github.com/LAMBA-23/LAMBA>



Deployed Swagger UI

<http://10.93.26.193:8000/docs>

